

Weaver Goldman

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EDUCATION

Worcester Polytechnic Institute

MS Data Science

BS Computer Science

Projected May 2027

GPA 4.0/4.0

Courses: MLOps, Machine Learning, Statistical Methods for ML, Database Management Systems, Algorithms

EXPERIENCE

AI Engineering Intern | Regeneron Pharmaceuticals, Inc.

May 2026 – Aug 2026

- Built an auditable AI/ML observability platform on [AWS](#) monitoring 30+ models, adopted by 3 pilot teams, and informed by 13 stakeholder interviews.
- Built a [Random Forest](#) drift detector using a classifier two-sample test to flag distribution shift against a baseline.
- Applied unsupervised [machine learning](#) (GMM, BIC, and cost matching) to track data subpopulations.
- Built a [React](#) dashboard to visualize model performance, data drift, and benchmarking results.
- Implemented an adaptive LLM benchmark system with over **60%** lower API costs than a naive evaluation strategy

Software Engineering Intern | Medtronic (Minimally Invasive Therapies Group)

Jun 2025 – Aug 2025

- Saved **400+** hours of manual work annually by automating surgical robot log analysis tasks with AI (LLMs).
- Developed an AI tool with [RAG](#) and [MCP](#) to analyze robot logs, performing expert-level analysis in minutes.
- Coordinated numerous AI agents and data sources with [LangChain/LangGraph](#) to perform complex tasks.

Web Development Lab Assistant | Laboratory of Spaceflight and Planetary Exploration

Aug 2024 – Present

- Under contract with [NASA](#), built a mission planning web app for exploring candidate trajectories to planets across the solar system.
- Optimized the database and graphing system so users can interactively filter **millions** of trajectories.
- Led development of an interactive 3D Mars lander simulation in [Three.js](#).
- Built a robust CI/CD pipeline for the lab with [Docker](#), unit testing, linting, and GitHub Actions.

Software Development and Research Intern | Gittlen Cancer Research Labs

May 2024 – Aug 2024

- Developed an [Electron](#) desktop application for extracting nerves and blood vessels from cloud-hosted medical scans, which reduces file sizes by up to **693×** (from 132.08 GB to 190.44 MB).
- Achieved approx. **40×** faster computation with multithreading and multiprocessing.
- Utilized [Docker](#) to create a cross-platform [Python](#) backend and an extensible plugin system.

PROJECTS

Scaling Diffusion Models to Large Sparse Graphs | PyTorch, Python

Aug 2025 – May 2026

- 4-person senior project on efficient diffusion architectures for molecule and protein generation in drug discovery.
- Achieved linear space complexity for our diffusion process on large, sparse graphs.

#TeamGraduate AI | Python, FastAPI, React, WordPress, Docker, Google Cloud Run

Jan 2026 – May 2026

- Developed a cloud-hosted LLM platform for generating curriculum-aligned (Namibian) educational content.
- Built AI authoring tools embedded in WordPress that reduced lesson creation time by **87.7%**.

Identifying Transit Accessibility Gaps in NYC | Scikit-Learn, Python, Plotly, GTFS

Apr 2025

- Identified underserved areas in NYC by using distance to the nearest subway station as a proxy for accessibility.
- Predicted weekly ridership revenue using a Random Forest Regressor based on **6** years of data.
- Visualized results with interactive geospatial maps, highlighting areas of opportunity based on accessibility and predicted revenue metrics (potential stops predicted to produce over **\$4 million** in revenue annually).

SKILLS

Languages: Python, SQL, JavaScript/TypeScript, C/C++, Java, L^AT_EX

AI & LLMs: LangChain/LangGraph, RAG, Model Context Protocol (MCP), agentic AI, embeddings, LLM evaluation

ML & Data Science: PyTorch, scikit-learn, TensorFlow, Pandas, NumPy, feature engineering, statistical modeling

MLOps & Cloud: AWS, GCP, Azure, Docker, CI/CD (GitHub Actions), Prometheus, Grafana, Weights & Biases

Data & Web: PostgreSQL, SQLite, DuckDB/Parquet, FastAPI, React, Angular, Streamlit, D3.js, Plotly Dash, Three.js

Tools & Practices: Git/GitHub, Linux, Bash, Figma, Jira